

EXPLANATIONS

1. (a) Potential at C due to charges q_1 and q_2 is

$$V_C = \frac{kq_1}{0.4} + \frac{kq_2}{0.5}$$

Potential at D due to charges q_1 and q_2 is

$$V_D = \frac{kq_1}{0.4} + \frac{kq_2}{0.1}$$

Change in potential energy is given by

$$\Delta U = q_3(V_D - V_C) = \left(\frac{kq_2}{0.1} - \frac{kq_2}{0.5} \right)$$

$$\Delta U = \frac{8q_2q_3}{4\pi\epsilon_0}$$

2. (a) Inside a charged spherical shell, potential equals the surface value,

$$V = \frac{kQ}{R} = 120 \text{ V at } R = 10 \text{ cm.}$$

At $r = 5 \text{ cm}$ (inside), $V = 120 \text{ V}$.

For $r = 15 \text{ cm}$ (outside),

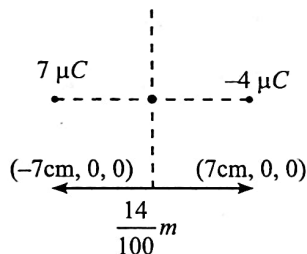
$$V = \frac{120 \times 10}{15} = 80 \text{ V.}$$

3. (b) Assertion (A) is true because no work is done in moving a charge in a region with zero electric field.

Reason (R) is true and correctly explains (A): The constant potential (due to zero field) implies no work is done.

Both A and R are true and R is the correct explanation of A.

4. (b)



The electrostatic potential energy is

$$U = \frac{Kq_1q_2}{r}$$

$$U = \frac{-9 \times 10^9 \times 4 \times 10^{-6} \times 7 \times 10^{-6}}{14} \times 100$$

$$U = -18 \times 10^{-1} \text{ J}$$

$$\boxed{U = -1.8 \text{ J}}$$

5. (c) Initial potential energy

$$= \frac{4Kq_0^2}{a} + \frac{2Kq_0^2}{\sqrt{2}a} = \frac{Kq_0^2}{a} (4 + \sqrt{2})$$

Final potential energy

$$= \frac{Kq_0^2}{\left(\frac{a}{\sqrt{2}}\right)} (4 + \sqrt{2}) = \frac{Kq_0^2}{a} (4\sqrt{2} + 2)$$

Change in potential energy

$$= \frac{Kq_0^2}{a} (3\sqrt{2} - 2)$$

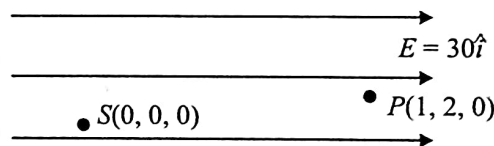
6. [18] The work done in rotating the electric dipole

$$\Delta U = U_f - U_i$$

$$\Delta U = (-pE \cos(180^\circ)) - (-pE \cos(0^\circ)) = 2pE$$

$$\Delta U = 2 \times 6 \times 10^{-6} \times 1.5 \times 10^3 = 18 \text{ mJ}$$

7. (c)



$$\therefore W = \vec{F} \cdot \vec{S}$$

$$\Rightarrow W_E = q\vec{E} \cdot \vec{S}$$

$$= 2 \times 10^{-2} [30\hat{i} \cdot (-\hat{i})]$$

$$= 2 \times 10^{-2} (-30)$$

$$\Rightarrow W = -600 \text{ mJ}$$

8. [243] As volume will same,

$$27 \times \frac{4}{3} \pi r^3 = \frac{4}{3} \pi R^3 \Rightarrow R = 3r$$

Using conservation of charge, $Q = 27q$

$$\text{Now potential energy, } = U_r = \frac{q^2}{8\pi\epsilon_0 r}$$

$$= U_R = \frac{Q^2}{8\pi\epsilon_0 R} = \frac{27 \times 27 \times q^2}{8\pi\epsilon_0 (3r)}$$

$$= 243 \left(\frac{q^2}{8\pi\epsilon_0 r} \right) = 243 U_r$$

Hence the potential energy of bigger drop is 243 times of smaller drop

9. (c) $F = -\frac{dU}{dr} = -\left[\frac{6.4 \cdot 12B}{r^2} - \frac{12B}{r^3}\right]$
 $F = 0$

$\Rightarrow r = \left(\frac{2B}{A}\right)^{1/2}$

$U\left(\text{at } r = \left(\frac{2B}{A}\right)^{1/2}\right) = \frac{A^2}{4B}$

10. (d) $\frac{1}{2}mv^2 = mgy + \Delta U_{\text{elec}}$

$\Delta U_{\text{elec}} = kQq \left[\frac{1}{R} - \frac{1}{(R+y)} \right]$

$\Rightarrow v^2 = 2y \left[\frac{qQ}{4\pi\epsilon_0 R(R+y)m} + g \right]$

11. (a) $U = -\frac{Kq^2}{d} + QV$

$V = \frac{KP}{D^2} = \frac{Kqd}{D^2}$

$U = -\frac{Kq^2}{d} - \frac{KQqd}{D^2}$

12. (a) $U = \frac{kq^2}{a} + \frac{kQq}{a} + \frac{kQq}{\sqrt{2}a} = 0$

$-q = Q \left(1 + \frac{1}{\sqrt{2}} \right)$

$Q = \frac{-q(\sqrt{2})}{\sqrt{2}+1}$

13. (b) $q_A = 1\mu\text{C}; q_B = 1\mu\text{C}; m_B = 4 \times 10^{-6} \text{ kg}$
 $r_{AB} = 10^{-3} \text{ m}$

$\frac{1}{2}Mv^2 = kq_Aq_B \left[\frac{1}{10^{-3}} - \frac{1}{9 \times 10^{-3}} \right]$

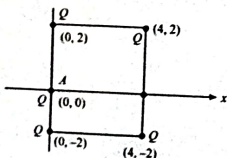
$\frac{1}{2} \times 4 \times 10^{-6} v^2 = 9 \times 10^9 \times 10^{-6} \times 10^{-6}$

$\times \frac{8}{9} \times 10^3$

$v^2 = \frac{8}{2} \times 10^6 = 4 \times 10^6$

$v = 2 \times 10^3 \text{ m/s}$

14. (d) $V_A = \frac{kQ}{2} + \frac{kQ}{2} + \frac{kQ}{\sqrt{2+4^2}} + \frac{kQ}{\sqrt{2+4^2}}$



$V_A = \frac{kQ}{2} + \frac{kQ}{2} + \frac{kQ}{\sqrt{2+4^2}} + \frac{kQ}{\sqrt{2+4^2}}$

$V_A = kQ + \frac{2kQ}{2\sqrt{5}}$

$PE = kQ^2 \left(1 + \frac{1}{\sqrt{5}} \right) = \frac{Q^2}{4\pi\epsilon_0} \left(1 + \frac{1}{\sqrt{5}} \right)$

15. (d) $\therefore \frac{Q^2}{4\pi\epsilon_0 R_0} = \frac{Q^2}{4\pi\epsilon_0 R} + \frac{1}{2}mv^2$

$\Rightarrow v = \sqrt{\frac{Q^2}{4\pi\epsilon_0} \times \frac{2}{m} \left[\frac{1}{R_0} - \frac{1}{R} \right]}$

So v increases and attains a finite value after large time.

16. [18] The work done in rotating the electric dipole $= \Delta U = U_f - U_i$
 $\Delta U = (-pE \cos(180^\circ)) - (-pE \cos(0^\circ)) = 2pE$
 $\Delta U = 2 \times 6 \times 10^{-4} \times 1.5 \times 10^3 = 18 \text{ mJ}$

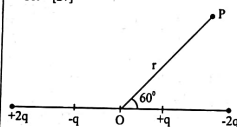
17. (c) $r_1 = \sqrt{(\sqrt{3})^2 + (\sqrt{3})^2}$
 $r_2 = \sqrt{(\sqrt{6})^2 + 0}$

As $r_1 = r_2 = \sqrt{6}$

Potential difference $= \frac{KQ}{r_1} - \frac{KQ}{r_2}$

So potential difference = 0

18. [27]

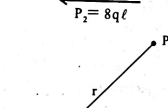


$\downarrow$

$P_1 = 2q\ell$

$\leftarrow$

$P_2 = 8q\ell$



$P_{\text{net}} = 6q\ell$

Potential energy,

$V = \frac{Kp \cdot \vec{r}}{r^3} = \frac{9 \times 10^9 (6q\ell)}{r^2} \cos(120^\circ)$
 $= -(27) \left(\frac{q\ell}{r^2} \right) \times 10^9 \text{ Nm}^2 \text{c}^{-2}$
 $\Rightarrow a = 27$

19. (b) As we know that,

$V = \frac{kP \cos \theta}{r^2}$

We can also check dimensionally,

20. [36] Potential $V = \int K \frac{dQ}{R} = \frac{KQ}{R}$

$V = \frac{K\lambda \pi R}{R} \Rightarrow V = K\lambda \pi$

$\Rightarrow V = 9 \times 10^9 \times 4 \times 10^{-4} \pi$

$\therefore V = 36\pi$

21. (b) Potential at the centre of the half ring

$V = \frac{(\lambda \cdot k\pi R)}{4\pi\epsilon_0 R_1} + \frac{(\lambda \cdot \pi R_1)}{4\pi\epsilon_0 R_1}$

$\Rightarrow V = \frac{\lambda}{2\epsilon_0}$

22. (c) Electric Potential $V = \frac{Kq}{r}$

$\therefore r = \frac{Kq}{V}$

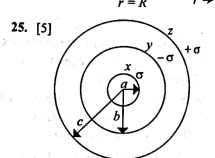
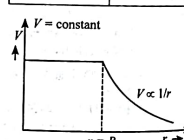
$r = \frac{9 \times 10^9 \times 5 \times 10^{-9}}{50}$

$r = \frac{45}{50} \times \frac{9}{10} = 0.9 \text{ m} = 90 \text{ cm}$

23. (c) Conceptual

24. (a)

Potential Inside	$V_{\text{in}} = \frac{KQ}{R}$
Potential at surface	$V_s = \frac{KQ}{R}$
Potential outside	$V_{\text{out}} = \frac{KQ}{r}$



Potential of X and Z are same.

$V_1 = V_2$
 $\frac{q_1}{4\pi\epsilon_0 a} + \frac{q_2}{4\pi\epsilon_0 b} + \frac{q_3}{4\pi\epsilon_0 c}$

$= \frac{q_1}{4\pi\epsilon_0 c} + \frac{q_2}{4\pi\epsilon_0 c} + \frac{q_3}{4\pi\epsilon_0 c}$

$\frac{q_1 a^2}{a} + \frac{q_2 b^2}{b} + \frac{q_3 c^2}{c}$

$= \frac{4\pi\epsilon_0 (a^2 - b^2 + c^2)}{c}$

$c(a-b+c) = a^2 - b^2$
 $c(a-b) + c(a-b) = a^2 - b^2$
 $c(a-b) = a^2 - b^2$
 $c(a-b) = (a-b)(a+b) \Rightarrow c = a+b=2$
 $c = 3 = 5 \text{ cm}$

26. [160] Let radius of smaller drop be r and that of bigger drop be R

Let q = charge on each drop

$\therefore V = \frac{Kq}{r}$

Now for combination of 64 drop

Volume of 64 drops = volume of single drop

$64 \times \frac{4}{3} \pi r^3 = \frac{4}{3} \pi R^3 \therefore R = 4r$

Charge on bigger drop $Q = 64q$

Potential of bigger drop

$V' = \frac{KQ}{R} = \frac{KQ}{4r} = 16 \frac{Kq}{r} = 16 \times 10 \text{ mV} = 160 \text{ mV}$

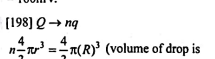
27. [198] $Q \rightarrow nq$
 $n \frac{4}{3} \pi r^3 = \frac{4}{3} \pi R^3$ (volume of drop is equal)

$\Rightarrow R = r n^{1/3}$

$V = \frac{kq}{r} \propto \frac{n}{n^{1/3}} \propto n^{2/3} \propto 27^{2/3}$ (potential calculated)

$\Rightarrow V' = 9V = 9 \times 22 = 198$

28. (d)



$Q_1 = q(4\pi R^2) = 4\pi R^2 q$
 $Q_2 = q(4\pi (2R)^2) = 16\pi R^2 q$

$Q_1' = \frac{Q_1}{4\pi\epsilon_0 R} = \frac{4\pi R^2 q}{4\pi\epsilon_0 R} = Rq$
 $Q_2' = \frac{Q_2}{4\pi\epsilon_0 (2R)} = \frac{16\pi R^2 q}{4\pi\epsilon_0 (2R)} = 2Rq$

$Q_1' + Q_2' = Rq + 2Rq = 3Rq$
 $\therefore Q_2' = 2Q_1'$

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$Q_1' + Q_2' = Q_1 + Q_2$

$\therefore \frac{Q_1}{2} + Q_2' = 20\pi R^2 q$

$\frac{3}{2} Q_2' = 20\pi R^2 q$

$\therefore \frac{Q_2'}{4\pi(2R)^2} = \frac{2}{3} \frac{20\pi R^2 q}{16\pi R^2}$

$\therefore \frac{Q_2'}{\sigma} = \frac{5}{6}$

29. (b) Let R = radius of combined drop, r = radius of smaller drop

Volume will remain constant,

$\frac{4}{3} \pi R^3 = 64 \times \frac{4}{3} \pi r^3$

$R = 4r$

q = charge of smaller drop

Q : Charge of combined drop

$\therefore Q = 64q$

From the formula of surface charge density

$\frac{Q}{4\pi R^2} = \frac{q}{4\pi r^2}$

$\frac{Q_{\text{bigger}}}{Q_{\text{smaller}}} = \frac{4\pi R^2}{4\pi r^2} = \frac{Q}{q} \cdot \frac{r^2}{R^2}$ (Ratio of)

surface density)

$= 64 \times \frac{r^2}{16r^2} = 4$

$\frac{Q_{\text{bigger}}}{Q_{\text{smaller}}} = 4$

30. (a) Potential at the centre of ring A, due to charge +Q,

$V_A = \frac{Q}{4\pi\epsilon_0} \left[\frac{1}{a} - \frac{1}{\sqrt{a^2 + s^2}} \right]$... (i)

Potential at the centre of ring B, due to charge -Q,

$V_B = \frac{-Q}{4\pi\epsilon_0} \left[\frac{1}{a} - \frac{1}{\sqrt{a^2 + s^2}} \right]$... (ii)

Subtracting (i) and (ii), we get,

$V_A - V_B = \frac{Q}{4\pi\epsilon_0} \left[\frac{1}{a} - \frac{1}{\sqrt{a^2 + s^2}} \right]$

31. [128] Step-1: Volume remains equal

Volume of small drop, $v = \frac{4}{3} \pi r^3$

Volume of large drop, $v = \frac{4}{3} \pi R^3$

Volume of 512 small drops

$= 512 \times \frac{4}{3} \pi r^3$

$512 \times \frac{4}{3} \pi r^3 = \frac{4}{3} \pi R^3$

$\Rightarrow R = 8r$

Step-2: Calculate the charge on large drop

Charge on large drop = 512 $\times$ charge on small drop

$Q = 512q$

Step-3: Calculate potential

Potential of small drop, $V = \frac{kq}{r} = 2V'$

Potential of large drop,

$V'' = \frac{kQ}{R} = \frac{k(512q)}{8r} = 64 \frac{kq}{r}$

$\Rightarrow V'' = 64 \times 2 = 128 V'$

Thus, the potential of the large drop is 128 V' .

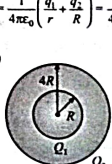
32. (a) Let charges on shells be q_1 and q_2
 $q_1 + q_2 = Q$... (i)

$\frac{q_1}{4\pi r^2} = \frac{q_2}{4\pi R^2}$... (ii)

We get $q_1 = \frac{r^2}{r^2 + R^2} Q$, $q_2 = \frac{R^2}{r^2 + R^2} Q$

$V = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{r} + \frac{q_2}{R} \right) = \frac{1}{4\pi\epsilon_0} \left(\frac{R}{r^2 + R^2} \right) Q$

33. (b)



$\therefore \frac{Q_1}{R^2} = \frac{Q_2}{16R^2}$

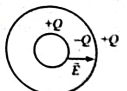
$\Rightarrow Q_2 = 16Q_1$

$\left(\frac{Q_1}{4R} + \frac{Q_2}{4R} \right) - \left(\frac{Q_1}{4R} + \frac{Q_2}{4R} \right) = V_1 - V_2$

$\Rightarrow V_1 - V_2 = \frac{3}{4} \times \frac{Q_1}{4\pi\epsilon_0 R} = \frac{3Q_1}{16\pi\epsilon_0 R}$

34. (b) In case-1

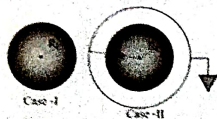
Electric field between spherical surface



In case-2

Electric field between surfaces remain unchanged.

64. (a)

Initially, $C_1 = 4\pi\epsilon_0 R_1$ and

$$\text{Finally, } C_2 = \frac{4\pi\epsilon_0 R_1 R_2}{R_1 - R_2}$$

According to question, $C_2 = nC_1$

$$\Rightarrow \frac{C_2}{C_1} = n$$

$$\frac{4\pi\epsilon_0 R_1 R_2}{R_1 - R_2} \times \frac{1}{4\pi\epsilon_0 R_1} = n$$

$$\frac{R_2}{R_1 - R_2} = n$$

$$nR_2 - nR_1 = R_2$$

$$(n-1)R_2 = nR_1$$

$$\Rightarrow \frac{R_2}{R_1} = \frac{n}{n-1}$$

65. (a) $E = \frac{\sigma}{\epsilon_0}$

$$F = qE = \frac{q\sigma}{\epsilon_0} = 10\text{ N}$$

On removing one plate, electric field will become half; therefore, force will also become half, i.e.,

$$F' = \frac{10}{2} = 5\text{ N}$$

66. (d) since force will act in horizontal direction so vertical component of speed will be same
-
- $v_i \cos \alpha = v_f \cos \beta$

$$\Rightarrow \frac{K_i}{K_f} = \left(\frac{v_i}{v_f}\right)^2 = \frac{\cos^2 \beta}{\cos^2 \alpha}$$

67. (a) $g_f = \sqrt{g^2 + \left(\frac{qE}{m}\right)^2}$

$$T = 2\pi \sqrt{\frac{L}{g_{eff}}} = 2\pi \sqrt{\frac{L}{g^2 + \left(\frac{qE}{m}\right)^2}}$$

68. (a) Current =
- $\frac{dq}{dt}$
- = slope of
- $q-t$
- graph

At $t = 4$ sec

Slope = 0

 $\therefore$ Current = 0

69. (d)
- $A = 30\pi \text{ cm}^2 = 30\pi \times 10^{-4} \text{ m}^2$

 $d = 1 \text{ mm} = 10^{-3} \text{ m}$ $E =$ dielectric strength for breakdown

$$= 3.6 \times 10^7 \text{ V/m}$$

$$Q = 7 \times 10^{-4} \text{ C}$$

$$\text{Electric field, } E = \frac{\sigma}{\epsilon_0}$$

$$E = \frac{Q}{kA\epsilon_0} \quad \therefore \sigma = \frac{Q}{A}$$

$$k = \frac{Q}{A\epsilon_0 E}$$

$$= \frac{7 \times 10^{-4} \times 4\pi \times 9 \times 10^9}{30\pi \times 10^{-4} \times 1 \times 3.6 \times 10^7}$$

$$= \frac{70}{30} = 2.33$$

70. (c) For
- $0 < x \leq \frac{d}{2}$

$$dC_1 = \frac{(\epsilon_0 + kx)A}{dx}$$

$$\int \frac{1}{dC_1} = \int \frac{dx}{(\epsilon_0 + kx)A}$$

$$\frac{1}{C_1} = \frac{1}{A} \frac{\ln(\epsilon_0 + kx)}{k}$$

$$= \frac{1}{kA} \ln\left(1 + \frac{kx}{\epsilon_0}\right)$$

$$C_1 = \frac{kA}{\ln\left(1 + \frac{kx}{\epsilon_0}\right)}$$

$$\text{Similarly, for } \frac{d}{2} < x \leq d,$$

$$dC_2 = \frac{[\epsilon_0 + k(d-x)]A}{dx}$$

$$\int \frac{1}{dC_2} = \int \frac{dx}{[\epsilon_0 + k(d-x)]A}$$

$$= \frac{1}{A} \frac{\ln[\epsilon_0 + k(d-x)]}{-k}$$

$$C_2 = \frac{kA}{\ln\left(1 + \frac{k(d-x)}{\epsilon_0}\right)}$$

$$\text{let, } C_1 = C_2 = C$$

$$\text{then equivalent capacitance}$$

$$= \frac{C_1 C_2}{C_1 + C_2} = \frac{C}{2} = \frac{kA}{2 \ln\left(1 + \frac{kx}{\epsilon_0}\right)}$$

71. [3] The given capacitor can be considered as two capacitor (one with dielectric and another without dielectric) connected in parallel.

$$\text{So, } \frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{d}{2k\epsilon_0 A} + \frac{d}{2\epsilon_0 A}$$

$$\Rightarrow C_{eq} = \frac{2\epsilon_0 A k}{d(k+1)} = \frac{2 \times 2 \times 3.2}{4.2 \times 1} \epsilon_0$$

$$= 3.04 \epsilon_0$$

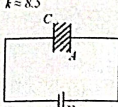
72. (d) $C = \frac{k\epsilon_0 A}{d}$

$$k = \frac{Cd}{\epsilon_0 A}$$

$$k = \frac{15 \times 10^{-12} \times 500 \times 10^{-6}}{8.86 \times 10^{-12} \times 10^{-4}}$$

$$= \frac{75}{8.86} = 8.465$$

$$k \approx 8.5$$



73. (c) $E_{net} = \frac{E_0}{k} + \frac{q_f}{\epsilon_0} + \frac{q_b}{\epsilon_0} = \frac{q_f}{k\epsilon_0}$

$$q_b = q_f \left(1 - \frac{1}{k}\right)$$

74. [864]

$$U_{eq} = \frac{C_1 V^2}{2} = \frac{14 \times 10^{-12} \times 12 \times 12}{2}$$

$$= 1008 \mu\text{J}$$

When a porcelain plate with $k = 7$ is inserted between the plates, the capacitance of system becomes $C = kC$

$$U_f = \frac{Q^2}{2C} = \frac{(CV)^2}{2kC} = \frac{CV^2}{2k}$$

$$= \frac{14 \times 10^{-12} \times 12 \times 12}{2 \times 7} = 144 \mu\text{J}$$

$$\text{Energy available for oscillation } U_a - U_f = 1008 - 144 = 864 \mu\text{J}$$

75. [161]

$$\frac{1}{C} = \frac{5}{\epsilon_0 \times 100} + \frac{5}{10 \times \epsilon_0 \times 100}$$

$$\Rightarrow C = \frac{8.85 \times 10^{-12} \times 1000}{55} = 161 \text{ pF}$$

76. (c) $E = \frac{\sigma}{\epsilon_0} = \frac{Q}{A\epsilon_0}$

$$Q = EA\epsilon_0 = 100 \times 1 \times 8.85 \times 10^{-12}$$

$$= 8.85 \times 10^{-10} \text{ C}$$

77. (b)

$$\frac{2\mu\text{C}}{3\mu\text{C} - 1\mu\text{C}} = \frac{4\mu\text{C}}{1\mu\text{C} - 3\mu\text{C}}$$

$$2\mu\text{C} = 4\mu\text{C} - 4\mu\text{C} = 0$$

$$2\mu\text{C} = 4\mu\text{C} - 4\mu\text{C} = 0$$

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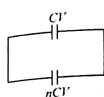
Charge at outer plates = $\frac{2+4}{2} = 3\mu\text{C}$

 $\therefore$ Charges at inner plates are $1\mu\text{C}$ and $-1\mu\text{C}$.

Potential difference = $\frac{q}{C} = \frac{1 \times 10^{-4} \text{ C}}{1 \times 10^{-4} \text{ F}}$

$$= 1 \text{ V}$$

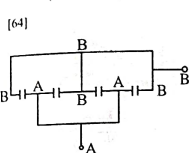
78. (a) Initially
- $Q = CV(1+n)$



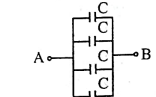
$$\therefore C_{eq} = (K+n)C$$

$$\therefore V = \frac{CV(1+n)}{(K+n)C} = \frac{V(1+n)}{(K+n)}$$

79. [64]



Redrawing the equivalent circuit



$$C_{eq} = 4C = 64$$

80. (b) When an uncharged capacitor C is connected in parallel to a charged capacitor
- $C_1 = 100\text{ pF}$
- with initial voltage
- $V_0 = 60\text{ V}$
- , the charge redistributes. The new potential is given by:

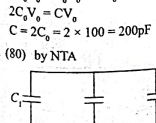
$$\text{New potential} = \frac{C_1 V_0}{C_0 + C} = \frac{V_0}{3}$$

Since the final voltage is 20 V , $V_f = 20$, so $V_0 = 60\text{ V}$ is consistent. The charge conservation

$$3C_1 V_0 = C_1 V_f + C V_f$$

$$2C_1 V_0 = C V_f = 200 \text{ pF}$$

81. (80) by NTA



$$C_1 = \frac{2\epsilon_0 A}{2 \times d} = 10\mu\text{F}$$

$$C_2 = \frac{3\epsilon_0 A}{2d} = 15\mu\text{F}$$

$$C_{eq} = C_1 + C_2$$

$$C_{eq} = 25\mu\text{F}$$

Charge on $C_1, Q_1 = 10\mu\text{C}$ Charge on $C_2, Q_2 = 1.5\text{ V}\mu\text{C}$

Now force between the plates

$$F = \frac{Q^2}{2A\epsilon_0}$$

$$F_1 + F_2 = F$$

$$\frac{100^2 \times 10^{-12}}{2 \times 2 \times 10^{-4} \times \epsilon_0} + \frac{225^2 \times 10^{-12}}{2 \times 2 \times 10^{-4} \times \epsilon_0} = 8$$

$$\frac{Q_1^2}{2A\epsilon_0} + \frac{Q_2^2}{2A\epsilon_0} = 8$$

$$V = 0.93 \times 10^{-3} \text{ V}$$

(So this must be a BONUS question)

82. (a) Capacitance of the capacitor,

$$\frac{A\epsilon_0}{d} = \frac{A\epsilon_0}{\left(0.2 + \frac{d}{k}\right)}$$

$$0.6 = 0.2 + \frac{0.6}{k}$$

$$\text{Dielectric constant of the capacitor,}$$

$$k = \frac{3}{2}$$

83. (a)
- $C_{eq} = C_1 + C_2 + C_3$

$$\Rightarrow C_{eq} = \frac{A\epsilon_0}{3d} + \frac{A\epsilon_0}{3(2d)} + \frac{A\epsilon_0}{3(3d)}$$

$$= \frac{A\epsilon_0}{3} \left(\frac{11}{6d}\right)$$

$$\therefore C_{eq} = \frac{11A\epsilon_0}{18d}$$

84. (c)
- $K_{metal} = \infty, \epsilon = \frac{2d}{3}$

$$C_1 = \frac{\epsilon_0 A}{d} = C$$

$$C_2 = \frac{\epsilon_0 A}{d-1} = \frac{\epsilon_0 A}{d - \frac{2d}{3}} = \frac{3\epsilon_0 A}{d}$$

$$= 3C_1 = 3C$$

85. [3] For $x = \frac{d}{3}$

$$C_1 = \frac{\epsilon_0 A}{\left(\frac{d}{3} + \frac{2d}{3}\right)} = \frac{\epsilon_0 A}{d}$$

$$= \frac{\epsilon_0 A}{d} \left(\frac{12}{9}\right)$$

$$C_1 = \frac{4\epsilon_0 A}{3d} = 2\mu\text{F} \Rightarrow \frac{\epsilon_0 A}{d} = \frac{4\mu\text{F}}{4}$$

$$\text{for } x = \frac{2d}{3}$$

$$C_2 = \frac{\epsilon_0 A}{\left(\frac{2d}{3} + \frac{d}{3}\right)} = \frac{\epsilon_0 A}{d} \times 2$$

$$= \frac{6}{4} \times 2 = 3\mu\text{F}$$

86. (a) Initial Capacitance $C = \frac{A\epsilon_0}{d}$

When sheet of thickness, $t = d/2$ is kept.Then $C_{new} = \frac{A\epsilon_0}{d-t} = \frac{A\epsilon_0}{d/2} = \frac{2A\epsilon_0}{d}$

$$\Rightarrow \frac{C_{new}}{C} = \frac{2}{1}$$

87. (c) Given, $C = \frac{\epsilon_0 A}{d} = 4\mu\text{F}$

Formula of capacitance, $C = \frac{\epsilon_0 A}{d}$

$$\therefore C_1 = \frac{\epsilon_0 A}{d} = 4 \times 2 = 8\mu\text{F}$$

$$C_2 = \frac{k\epsilon_0 A}{d/2} = 3 \times 2 \times 4 = 24\mu\text{F}$$

Since, C_1 and C_2 are in series, therefore,

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} = \frac{8 \times 24}{8 + 24} = \frac{8 \times 24}{32} = 6\mu\text{F}$$

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} = \frac{8 \times 24}{8 + 24} = 6\mu\text{F}$$

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} = \frac{8 \times 24}{8 + 24} = 6\mu\text{F}$$

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$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} = \frac{8 \times 24}{8 + 24} = 6\mu\text{F}$$

88. [23] Capacitance of parallel plate capacitor

$$C = \frac{\epsilon_0 A}{d}$$

Where, $d =$ distance between the plates of capacitor.

$$C_1 = \frac{\epsilon_0 A}{3b}, C_2 = \frac{\epsilon_0 A}{3b}, C_3 = \frac{\epsilon_0 A}{b}$$

Since, capacitors are connected in parallel therefore,

$$C_{\text{eq}} = C_1 + C_2 + C_3 = \frac{\epsilon_0 A}{b} \left[\frac{1}{5} + 1 + 1 \right]$$

$$C_{\text{eq}} = \frac{23\epsilon_0 A}{15b}$$

89. (*) For series combination of capacitors of capacitance C_1 and C_2 , $C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2}$

$$\frac{C_1 \times C_2}{C_1 + C_2} = \frac{15}{4} (C_1 + C_2) \Rightarrow 4C_1 C_2 = 15(C_1^2 + C_2^2 + 2C_1 C_2)$$

$$\Rightarrow 4 \left(\frac{C_1}{C_2} \right) = 15 \left(\frac{C_1}{C_2} \right)^2 + 30 \left(\frac{C_1}{C_2} \right) + 15 \Rightarrow 15x^2 + 26x + 15 = 0$$

Where, $x = \frac{C_1}{C_2}$

$$\Rightarrow x = \frac{-26 \pm \sqrt{676 - 900}}{30}$$

Since x cannot be real, so this Question has been dropped by NTA.

90. (a) Capacitors, C_1 , C_2 and C_3 are in series therefore,

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$\frac{1}{C_{\text{eq}}} = \frac{1}{\left(\frac{k\epsilon_0 A}{d} \right)} + \frac{1}{\left(\frac{3k\epsilon_0 A}{2d} \right)} + \frac{1}{\left(\frac{5k\epsilon_0 A}{3d} \right)}$$

$$\frac{1}{C_{\text{eq}}} = \frac{d}{k\epsilon_0 A} + \frac{2d}{3k\epsilon_0 A} + \frac{3d}{5k\epsilon_0 A}$$

$$\frac{1}{C_{\text{eq}}} = \frac{(15+10+9)d}{15k\epsilon_0 A}$$

$$C_{\text{eq}} = \frac{15k\epsilon_0 A}{34d}$$

91. (d) $T \sin \theta = qE$

$$T \cos \theta = mg$$

On dividing (i) by (ii) we get,

$$\tan \theta = \frac{qE}{mg}$$

$$Q = \frac{C_1 C_2}{C_1 + C_2} (V_1 + V_2)$$

$$E = \frac{Q}{A\epsilon_0} = \frac{C_1 C_2}{C_1 + C_2} \left[\frac{V_1 + V_2}{A\epsilon_0} \right]$$

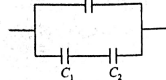
$$C_1 = \frac{\epsilon_0 A}{d - t} = \frac{C_2 (V_1 + V_2)}{(C_1 + C_2)(d - t)}$$

$$\theta = \tan^{-1} \left[\frac{qE}{mg} \right]$$

$$\Rightarrow \theta = \tan^{-1} \left[\frac{q}{mg} \times \frac{C_2 (V_1 + V_2)}{(C_1 + C_2)(d - t)} \right]$$

92. (c) Here, $C = \frac{\epsilon_0 A}{d}$, $C_1 = \frac{k_1 \epsilon_0 A}{2d} = \frac{K_1 \epsilon_0 A}{d}$

$$\text{and } C_2 = \frac{k_2 \epsilon_0 A}{2d} = \frac{K_2 \epsilon_0 A}{d}$$



$$C_{\text{eq}} = C + \frac{C_1 \times C_2}{C_1 + C_2}$$

On putting the value of C , C_1 and C_2 in the above equation, we get,

$$C_{\text{eq}} = \frac{\epsilon_0 A}{d} \left(1 + \frac{k_1 k_2}{k_1 + k_2} \right)$$

93. (a) Capacitance of element, $d =$

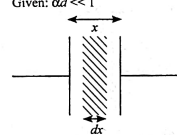
$$\frac{K(1 + \alpha x) \epsilon_0 A}{dx}$$

As all elements are in series, so equivalent capacitance

$$\sum \frac{1}{C} = \int \frac{1}{dC} = \int_0^d \frac{dx}{k\epsilon_0 A(1 + \alpha x)}$$

$$\frac{1}{C} = \frac{1}{k\epsilon_0 A} \ln \left(\frac{1 + \alpha d}{1} \right)$$

Given: $\alpha d \ll 1$

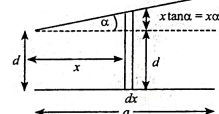


$$\frac{1}{C} = \frac{1}{K\epsilon_0 A d} \left(\alpha d - \frac{\alpha^2 d^2}{2} \right)$$

$$\Rightarrow \frac{1}{C} = \frac{d}{K\epsilon_0 A} \left(1 - \frac{\alpha d}{2} \right)$$

$$\Rightarrow C = \frac{AK\epsilon_0 A}{d} \left(1 + \frac{\alpha d}{2} \right)$$

94. (d)



We know that, $C = \frac{\epsilon_0 A}{d}$

$$dC = \frac{\epsilon_0 A dx}{d^2} \Rightarrow \int dC = \int \frac{\epsilon_0 A dx}{d^2}$$

$$\Rightarrow C = \frac{\epsilon_0 A}{d} \ln \left(\frac{1 + \alpha d}{1} \right) = \frac{\epsilon_0 A}{d} \left(1 - \frac{\alpha d}{2} \right)$$

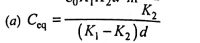
$$\Rightarrow C = \frac{\epsilon_0 A}{d} \ln \left(1 + \frac{\alpha d}{1} \right) = \frac{\epsilon_0 A}{d} \left(1 - \frac{\alpha d}{2} \right)$$

$$95. (c) C = C_1 + C_2 + C_3$$

$$\frac{\epsilon_0 K A}{d} = \frac{\epsilon_0 K_1 A}{3d} + \frac{\epsilon_0 K_2 A}{3d} + \frac{\epsilon_0 K_3 A}{3d}$$

$$\Rightarrow K = \frac{K_1 + K_2 + K_3}{3} = \frac{10 + 12 + 14}{3} = 12$$

$$96. (a) C_{\text{eq}} = \frac{\epsilon_0 K_1 K_2 a^2 \ln \frac{K_1}{K_2}}{(K_1 - K_2)d}$$



$$\Rightarrow C_{\text{eq}} = \frac{\epsilon_0 K a^2 \ln \frac{1}{1-K}}{d(1-K)}$$

$$\text{As } K_1 = 1, K_2 = K$$

$$\Rightarrow C_{\text{eq}} = \frac{\epsilon_0 K a^2}{d(K-1)} \ln K$$

$$97. (*) C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2} + \frac{C_3 C_4}{C_3 + C_4}, C_1 = K_1 C$$

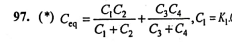
$$C = \frac{\epsilon_0 A}{d} \ln 2$$

$$\text{Similarly } C_2 = K_2 C, C_3 = K_3 C, C_4 = K_4 C$$

$$C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2} + \frac{C_3 C_4}{C_3 + C_4} = \frac{K_1 K_2}{K_1 + K_2} \frac{\epsilon_0 A}{d} \ln 2$$

$$\Rightarrow C_{\text{eq}} = \frac{K_1 K_2}{K_1 + K_2} \frac{\epsilon_0 A}{d} \ln 2$$

$$98. (c) \text{ Reduced circuit is,}$$



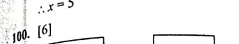
$$\text{Both capacitors are in parallel.}$$

$$\therefore C_{\text{total}} = C + C = 2C$$

$$99. [5] C = \frac{\epsilon_0 A}{(d - t)}$$

$$= \frac{\epsilon_0 \times 200 \times 10^{-4}}{4 \times 10^{-3}} = 5 \epsilon_0$$

$$\therefore x = 5$$



$$100. [6] C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2} = \frac{24}{10} A\epsilon_0$$

$$q_{\text{in}} = C_{\text{eq}} V = 240 A\epsilon_0$$

$$\Delta V_2 = \frac{240 A\epsilon_0}{4 A\epsilon_0} = 60V$$

$$V_{\text{in}} = 60V$$

$$105. (d) C_1 = \frac{\epsilon_0 A}{3d/4} = \frac{4\epsilon_0 A}{3d} = \frac{4K C_0}{3}$$

$$C_2 = \frac{\epsilon_0 A}{d/4} = \frac{4\epsilon_0 A}{d} = 4C_0$$

$$\Rightarrow 40 K = 24 + 24 K$$

$$\Rightarrow K = \frac{24}{16} = 1.5$$

$$101. (a) C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2} = \frac{40 \times 40 K}{40(1 + K)} = 24$$

$$\Rightarrow 40 K = 24 + 24 K$$

$$\Rightarrow K = \frac{24}{16} = 1.5$$

$$102. [12] C_{\text{eq}} = 4 + 4 + 4 = 12 \mu F$$

$$103. (a) \text{ We know that,}$$

$$C_0 = \frac{\epsilon_0 A}{d}$$

$$C_1 = \frac{4\epsilon_0 A}{d} = 4C_0$$

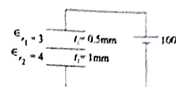
$$C_2 = \frac{K\epsilon_0 A}{3d} = \frac{4K C_0}{3}$$

$$\text{Series,}$$

$$C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2} = \frac{4K C_0 \cdot 3}{3(K + 3)} = \frac{4K C_0}{K + 3}$$

$$104. [60] \text{ Capacitance of each capacitor}$$

$$C_1 = \epsilon_0 \frac{A}{t} = 6 A\epsilon_0, C_2 = \frac{\epsilon_0 A}{t} = 4 A\epsilon_0$$



Equivalent capacitance

$$C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2} = \frac{24}{10} A\epsilon_0$$

$$q_{\text{in}} = C_{\text{eq}} V = 240 A\epsilon_0$$

$$\Delta V_2 = \frac{240 A\epsilon_0}{4 A\epsilon_0} = 60V$$

$$V_{\text{in}} = 60V$$

$$105. (d) C_1 = \frac{\epsilon_0 A}{3d/4} = \frac{4\epsilon_0 A}{3d} = \frac{4K C_0}{3}$$

$$C_2 = \frac{\epsilon_0 A}{d/4} = \frac{4\epsilon_0 A}{d} = 4C_0$$

$$\Rightarrow 40 K = 24 + 24 K$$

$$\Rightarrow K = \frac{24}{16} = 1.5$$

$$106. (d) \text{ Formula of common potential}$$

$$V_c = \frac{Q_1 + Q_2}{C_1 + C_2} = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

$$V_c = \frac{2C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

$$V_c = \frac{3C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

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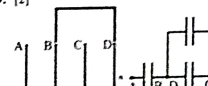
$$\Rightarrow 4 \left(\frac{C_1}{C_2} \right) = 15 \left(\frac{C_1}{C_2} \right)^2 + 30 \left(\frac{C_1}{C_2} \right) + 15 \Rightarrow 15x^2 + 26x + 15 = 0$$

Where, $x = \frac{C_1}{C_2}$

$$\Rightarrow x = \frac{-26 \pm \sqrt{676 - 900}}{30}$$

Since x cannot be real, so this Question has been dropped by NTA. Its is bonus.

109. [2]



$$C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2} = \frac{24}{10} A\epsilon_0$$

$$q_{\text{in}} = C_{\text{eq}} V = 240 A\epsilon_0$$

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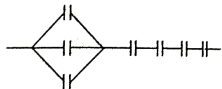
$$\frac{C_1 C_2}{C_1 + C_2} = \frac{80}{10} = 8 \mu\text{F}$$

$$\therefore C_1 C_2 = 400 \mu\text{F}$$

$$\text{Solving } C_1 = 40 \mu\text{F}, C_2 = 10 \mu\text{F}$$

$$112. (b) C_{eq} = \frac{6}{13} \mu\text{F}$$

Therefore three capacitors must be in parallel to get 6 in



$$\frac{1}{C_{eq}} = \frac{1}{3C} + \frac{1}{C} + \frac{1}{C} + \frac{1}{C} + \frac{1}{C}$$

$$C_{eq} = \frac{3C}{13} = \frac{6}{13} \mu\text{F}$$

$$113. (c) \frac{1}{C_1} = \frac{d}{3A\epsilon_0} \left(\frac{1}{K_1} + \frac{1}{K_2} + \frac{1}{K_3} \right)$$

$$C_1 = \frac{3A\epsilon_0 (K_1 K_2 K_3)}{d(K_1 K_2 + K_2 K_3 + K_3 K_1)}$$

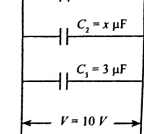
$$C_2 = \frac{A\epsilon_0}{3d} (K_1 + K_2 + K_3)$$

$$\frac{E_1}{E_2} = \frac{C_1}{C_2} = \frac{3K_1 K_2 K_3}{(K_1 K_2 + K_2 K_3 + K_3 K_1)}$$

$$\times \frac{3}{(K_1 + K_2 + K_3)}$$

$$\Rightarrow \frac{E_1}{E_2} = \frac{9K_1 K_2 K_3}{(K_1 + K_2 + K_3)(K_1 K_2 + K_2 K_3 + K_3 K_1)}$$

$$114. [5] \quad C_1 = 2 \mu\text{F}$$



$$\text{Charge on } C_1 \text{ is } Q_1 = C_1 V = 2 \times 10 = 20 \mu\text{C}$$

$$\text{Charge on } C_2 \text{ is } Q_2 = C_2 V = 10x \mu\text{C}$$

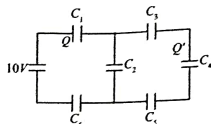
$$\text{Charge on } C_3 \text{ is } Q_3 = C_3 V = 30 \mu\text{C}$$

$$\text{Total charge } Q = Q_1 + Q_2 + Q_3$$

$$100 = 20 + 10x + 30$$

$$\Rightarrow x = 5$$

$$115. [4]$$



C_1, C_2 and C_5 are in series

$$\therefore \frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_5}$$

$$= \frac{1}{2} + \frac{1}{4} + \frac{1}{2} \Rightarrow C = \frac{4}{5} = 0.8 \mu\text{F}$$

Now, C and C_3 are in parallel, therefore,

$$C' = C + C_3 = 0.8 + 0.2 = 1 \mu\text{F}$$

Now, C', C_4 and C_6 are in series.

$$\frac{1}{C_{eq}} = \frac{1}{C'} + \frac{1}{C_4} + \frac{1}{C_6} = \frac{1}{1} + \frac{1}{2} + \frac{1}{2}$$

$$= 2 \Rightarrow C_{eq} = 0.5 \mu\text{F}$$

$$Q = 0.5 \times 10 = 5 \mu\text{C}$$

$$Q' = \frac{5 \mu\text{C} \times 0.8}{0.8 + 0.2} = 4 \mu\text{C}$$

$$116. (a) \text{ Equivalent capacitance of the system}$$

$$C_{eq} = C_1 + C_2 + C_3 + C_4$$

$$= 1 + 2 + 4 + 3 = 10 \mu\text{F}$$

Charge,

$$Q = C_{eq} V$$

$$= (10 \times 20) \mu\text{C}$$

$$Q = 200 \mu\text{C}$$

$$117. (a) \text{ Equivalent capacitance,}$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \frac{1}{C_4} + \frac{1}{C_5} + \frac{1}{C_6}$$

$$= \frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \frac{1}{3} + \frac{1}{5} + \frac{1}{20}$$

$$\frac{1}{C_{eq}} = \frac{6 + 4 + 3}{60} = \frac{13}{60}$$

$$C_{eq} = \frac{60}{13} \mu\text{F}$$

Total charge in the circuit

$$Q = C_{eq} V = \frac{60}{13} \times 13 \mu\text{C} = 60 \mu\text{C}$$

$$118. (d) \text{ We have given, } C_1 = 2 \mu\text{F}, C_2 = 6 \mu\text{F}$$

$$\text{and } C_3 = 12 \mu\text{F}$$

$$Q_1 = C_1 V = 2V \mu\text{C}$$

$$Q_2 = Q_1 = \frac{C_1 C_2}{C_1 + C_2} V = \frac{6 \times 12}{6 + 12} V = 4V \mu\text{C}$$

$$Q_3 = Q_2 = \frac{C_2 C_3}{C_2 + C_3} V = \frac{12 \times 12}{12 + 12} V = 6V \mu\text{C}$$

$$Q_4 = Q_3 = \frac{C_3 C_4}{C_3 + C_4} V = \frac{12 \times 12}{12 + 12} V = 6V \mu\text{C}$$

$$Q_5 = Q_4 = \frac{C_4 C_5}{C_4 + C_5} V = \frac{12 \times 12}{12 + 12} V = 6V \mu\text{C}$$

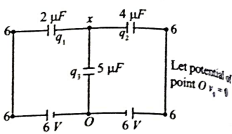
$$Q_6 = Q_5 = \frac{C_5 C_6}{C_5 + C_6} V = \frac{12 \times 12}{12 + 12} V = 6V \mu\text{C}$$

$$Q_7 = Q_6 = \frac{C_6 C_7}{C_6 + C_7} V = \frac{12 \times 12}{12 + 12} V = 6V \mu\text{C}$$

$$Q_8 = Q_7 = \frac{C_7 C_8}{C_7 + C_8} V = \frac{12 \times 12}{12 + 12} V = 6V \mu\text{C}$$

$$119. (a) \text{ Using junction analysis}$$

$$\text{We can say, } q_1 + q_2 + q_3 = 0$$



We know $Q = CV$ [Where, C is capacitance and V is voltage across the capacitor]

$$2(x - 6) + 4(x - 6) + 5(x) = 0$$

$$x = \frac{36}{11} \Rightarrow q_1 = \frac{36(5)}{11} = \frac{180}{11}$$

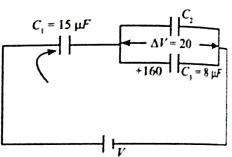
$$q_2 = 16.36 \mu\text{C}$$

$$120. (c) \text{ As potential difference across } C_1 \text{ and } C_2 \text{ are equal}$$

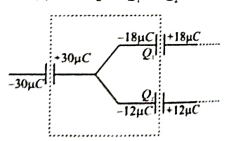
$$\text{So } Q_1 = C_1 V = 160 \mu\text{C}$$

$$\therefore Q_2 = 750 \mu\text{C} - 160 \mu\text{C} = 590 \mu\text{C}$$

$$Q_1 = C_1 V, Q_2 = C_2 V$$



$$121. (a) \text{ Let charge be } Q_1 \text{ and } Q_2$$



$$\frac{Q_1}{6} = \frac{Q_2}{4}$$

$$\Rightarrow Q_1 = Q_2 = 30$$

$$\Rightarrow Q_1 = 18 \mu\text{C}, Q_2 = 12 \mu\text{C}$$

$$122. (b) \quad 4 \mu\text{F}, 6 \mu\text{F}, 3 \mu\text{F}, 2.4 \mu\text{F}, 3 \mu\text{F}, 10 \mu\text{F}$$

$$\text{Total charge } Q = 5.4 \mu\text{F} \times 10 \text{ V} = 54 \mu\text{C}$$

$$\text{The charge in branch } PQ, Q_{PQ}$$

$$= 2.4 \mu\text{F} \times 10 \text{ V} = 24 \mu\text{C}$$

$$\text{Since } 4 \mu\text{F} \text{ is in series with } 6 \mu\text{F}$$

$$(= 1 \mu\text{F}) [5 \mu\text{F}], \text{ So } Q_{6 \mu\text{F}} = 24 \mu\text{C}$$

$$123. (c) \text{ A mobile phone battery is rated at } 4.2 \text{ V and } 5800 \text{ mAh. Energy stored is calculated as } E = V \times Q, \text{ where } Q = 5800 \text{ mAh} = 5800 \times 3.6 \times 10^3 \text{ J}$$

$$\text{Thus, } E = 4.2 \times 5800 \times 3.6 \times 10^3 = 87,696 \text{ kJ} = 87.7 \text{ kJ}$$

$$124. (a) \Delta q = (KC - C)V$$

$$= 40 \times 10^{-4} \times 100 = 4 \text{ mC}$$

$$\Delta U = \frac{1}{2} C V^2 - \frac{1}{2} C V^2 = \frac{1}{2} (K - 1) C V^2$$

$$= \frac{1}{2} C V^2 (2 - 1) = \frac{1}{2} C V^2$$

$$= \frac{1}{2} \times 40 \times 10^{-4} \times 10000 = 0.2 \text{ J}$$

$$125. (e) \text{ Energy per unit volume is given by:}$$

$$\frac{U}{V} = \frac{1}{2} \epsilon_0 E^2$$

$$\text{Since volume } V \text{ of the capacitor is } A \times d, \text{ the total energy stored in the capacitor is:}$$

$$U = \frac{1}{2} \epsilon_0 E^2 \times A d$$

$$126. (c) (Q_1)_{\text{max}} < (Q_2)_{\text{max}}$$

$$C_1 V < C_2 V$$

$$C_1 > C_2$$

$$U = \frac{1}{2} C V^2$$

$$\Rightarrow U < C$$

$$\Rightarrow U_2 > U_1$$

$$127. (b) C = 1 \mu\text{F}, V = 20 \text{ V}, d = 1 \text{ mm}$$

$$\text{So, } E = \frac{V}{d} = 20 \times 10^3 \text{ V/m}$$

$$\text{Energy density is given by } \frac{1}{2} \epsilon_0 E^2$$

$$U = 1.77 \times 10^3 \text{ J/m}^3 = 1.8 \times 10^3 \text{ J/m}^3$$

$$128. (c) \text{ Initially, capacitor } C_1 = 6 \mu\text{F} \text{ is charged to a potential difference } V_1 = 5 \text{ V using a } 5 \text{ V battery. The charge on } C_1 \text{ is:}$$

$$q_1 = C_1 V_1 = 6 \times 5 = 30 \mu\text{C}$$

$$\text{Finally,}$$

$$q_1 = 6V_1, q_2 = 12V_2$$

$$q_1 = -6V_1, q_2 = -12V_2$$

$$\text{The battery is then removed, and capacitor } C_2 = 12 \mu\text{F} \text{ is inserted in place of the battery. The switch } S \text{ is closed, and charge flows between the capacitors until equilibrium is reached.}$$

$$\text{At equilibrium, the total charge is shared between } C_1 \text{ and } C_2, \text{ and they both have the same potential difference } V. \text{ We can write the equation for the total charge:}$$

$$6V_1 + 12V_2 = 30$$

$$18V_1 = 30 \Rightarrow V_1 = \frac{30}{18} = \frac{5}{3} \text{ V}$$

Final charges:

$$q_1 = C_1 \times V_1 = 6 \times \frac{5}{3} = 10 \mu\text{C}$$

$$q_2 = C_2 \times V_2 = 12 \times \frac{5}{3} = 20 \mu\text{C}$$

$$129. [2] \text{ As we know that,}$$

$$\text{Energy loss} = \frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} (V_1 - V_2)^2$$

$$\text{Hence, Energy loss} = \frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} (V_1 - V_2)^2$$

$$\therefore x = 2$$

$$130. (a) C_{eq} = 2C$$

$$V_C = \frac{q_{net}}{C_{net}} = \frac{CV + 2CV}{2C} = \frac{3V}{2}$$

Therefore, decrease in energy of the combined system,

$$= \frac{1}{2} C V^2 - \frac{1}{2} C (2V)^2 = -\frac{1}{2} C (3V)^2$$

$$= -\frac{1}{2} C (3V)^2 = -\frac{1}{2} C (3V)^2$$

$$= -\frac{1}{2} C (3V)^2 = -\frac{1}{2} C (3V)^2$$

$$131. [86] \text{ In parallel combination: Potential difference is same across all capacitors } = V = 100 \text{ V}$$

$$U = \frac{1}{2} C V^2$$

$$U = \frac{1}{2} (C_1 + C_2 + C_3) V^2$$

$$= \frac{1}{2} (25 + 30 + 45) \times (100)^2 \times 10^{-6}$$

$$= 0.5 \text{ J}$$

In series combination: Charge is same on all capacitors.

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$= \frac{1}{25} + \frac{1}{30} + \frac{1}{45} = C = 450$$

$$= 125 \times 10^{-6} \text{ J} = 125 \mu\text{J}$$

$$136. (c) \text{ Given: } C_1 + C_2 = 10 \mu\text{F}$$

$$\text{Energy stored in the capacitor} = \frac{1}{2} C V^2$$

$$\text{So, according to given condition}$$

$$4 \left(\frac{1}{2} C V^2 \right) = \frac{1}{2} C V^2$$

$$\Rightarrow 4C_1 = C_2$$

$$\text{From equation (i) and (ii)}$$

$$C_1 = 2 \mu\text{F}$$

$$C_2 = 8 \mu\text{F}$$

$$C_{eq} \text{ in series} = \frac{C_1 C_2}{C_1 + C_2} = 1.6 \mu\text{F}$$

$$132. (c) \text{ We have given that the two charged conducting spheres of radii } a \text{ and } b \text{ are connected to each other by a conducting wire.}$$

$$\text{Their potential will be equal}$$

$$\frac{Kq_1}{a} = \frac{Kq_2}{b}$$

$$\frac{q_1}{a} = \frac{q_2}{b}$$

$$133. [55] \text{ As source is connected in the process therefore, additional charge will be provided. Therefore, charge on } C_1 = K(CV).$$

Since source is disconnected, so its charge will remain the same. Then, Charge on $C_2 = CV$.

New capacitance in both cases will become, KC .

When they are connected in parallel charge will be equally divided so charge on one capacitor is,

$$q = \left(\frac{K+1}{2} \right) CV$$

So common potential,

$$V' = \frac{q}{KC} = \frac{K+1}{2K} V = 55V$$

$$134. [4] \text{ Battery is connected while dielectric is inserted so potential difference will be remains same}$$

$$U_1 = \frac{1}{2} C V^2$$

$$U_2 = \frac{1}{2} K C V^2$$

$$\Rightarrow \Delta U = \left(\frac{1}{2} K - 1 \right) C V^2$$

$$= \frac{1}{2} \times 1 \times 200 \times 10^{-6} \times 200^2 = 4$$

$$135. [125] \text{ Electrostatic Energy loss,}$$

$$= \frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} (V_1 - V_2)^2$$

$$= \frac{1}{2} \frac{50 \times 10^{-12} \times 50 \times 10^{-12}}{50 \times 10^{-12} + 50 \times 10^{-12}} \times (100 - 0)^2$$

$$= 125 \times 10^{-6} \text{ J} = 125 \mu\text{J}$$

$$136. (c) \text{ Given: } C_1 + C_2 = 10 \mu\text{F}$$

$$\text{Energy stored in the capacitor} = \frac{1}{2} C V^2$$

$$\text{So, according to given condition}$$

$$4 \left(\frac{1}{2} C V^2 \right) = \frac{1}{2} C V^2$$

$$\Rightarrow 4C_1 = C_2$$

$$\text{From equation (i) and (ii)}$$

$$C_1 = 2 \mu\text{F}$$

$$C_2 = 8 \mu\text{F}$$

$$C_{eq} \text{ in series} = \frac{C_1 C_2}{C_1 + C_2} = 1.6 \mu\text{F}$$

17. (b) Initially, Charge on capacitor

$$Q_i = CV = 2V$$

Energy stored

$$E_i = \frac{Q_i^2}{2C} = \frac{4V^2}{2 \times 2} = V^2$$

$$E_f = 1/2 CV^2 = 1/2 (2) V^2 = V^2$$

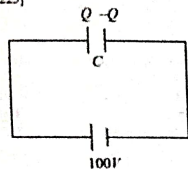
Finally

Charge on each capacitor,

$$Q_1 = \frac{Q_i}{2} = \frac{2V}{2} = V$$

$$E_1 = 2 \left(\frac{1}{2} \frac{Q_1^2}{C} \right) = \frac{V^2}{2} \therefore \frac{E_1}{E_i} = \frac{1}{2}$$

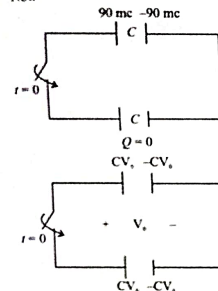
138. [223]



$$C = 900 \mu F$$

$$Q = CV = 900 \times 10^{-6} \times 100 = 9 \times 10^{-2} = 90 \text{ mC}$$

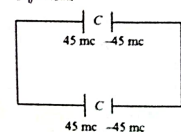
Now



Common potential will be developed across both capacitors by kVZ. Total charge on left plates of capacitors should be conserved.

$$\therefore 90 \text{ mC} + 0 = 2Cv_0$$

$$Cv_0 = 45 \text{ mC}$$



Heat dissipated = $U_i - U_f$ [Change in energy stored in the capacitors]

$$= \frac{1}{2} \frac{(90 \text{ mC})^2}{900 \mu F} - 2 \times \frac{1}{2} \frac{(45 \text{ mC})^2}{900 \mu F} \left[U = \frac{Q^2}{2C} \right]$$

$$= \frac{1}{2 \times 900 \times 10^{-6}} (8100 - 4050) \times 10^{-6}$$

$$= 2.25 \text{ Joule}$$

OR

$$\text{Heat} = \frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} (V_1 - V_2)^2$$

$$= \frac{1}{2} \frac{C^2}{2C} (100 - 0)^2$$

$$= \frac{1}{2} \frac{900 \times 10^{-6}}{2} \times 10^4 = \frac{9}{4}$$

$$\text{Joule} = 2.25 \text{ Joule}$$

139. [6] Formula of electrostatic loss of energy

$$\text{Energy lost} = \frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} (V_1 - V_2)^2$$

$$= \frac{1}{2} \times \frac{(600 \times 600)}{(600 + 600)} \times (200)^2$$

$$= 6 \times 10^6 \times 10^{-12} = 2 \times 10^{-4} \text{ J} = 6 \mu \text{J}$$

140. [125] Electrostatic Energy loss,

$$= \frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} (V_1 - V_2)^2$$

$$= \frac{1}{2} \times \frac{50 \times 10^{-12} \times 50 \times 10^{-12}}{50 \times 10^{-12} + 50 \times 10^{-12}} \times (100 - 0)^2$$

$$= 125 \times 10^{-4} \text{ J}$$

$$= 125 \text{ nJ}$$

141. (a) Let us suppose,

$$\text{Initial charge on capacitor, } q_1 = q$$

$$\text{Final charge on capacitor, } q_2 = q + 2$$

According to the question,

$$\frac{U_2 - U_1}{U_1} = \frac{(q+2)^2 - q^2}{q^2} = \frac{44}{100}$$

$$\Rightarrow \frac{(q+2)^2 - q^2}{q^2} = \frac{44}{100}$$

$$\Rightarrow \left(1 + \frac{2}{q}\right)^2 - 1 = \frac{44}{100} \Rightarrow \left(1 + \frac{2}{q}\right)^2 = \frac{144}{100}$$

$$\Rightarrow \left(1 + \frac{2}{q}\right)^2 = \frac{12}{10} \Rightarrow \frac{2}{q} = 0.2 \Rightarrow q = 10 \text{ C}$$

142. (c) Case (I): When switch is closed, both capacitors are in parallel and energy stored. $C_{eq} = 2C$

$$\text{Energy, } E_1 = \frac{1}{2} C_{eq} V^2 = CV^2$$

Case (II): When switch is opened, one charge get disconnected.

$$E_2 = \frac{1}{2} \times (5C) V^2 + \frac{(CV)^2}{2 \times 5C} = \frac{13}{5} CV^2$$

$$\therefore \text{Required ratio, } \frac{E_1}{E_2} = \frac{5}{13}$$

$$143. (a) U = \frac{1}{2} CV^2$$

It is given that, voltage is constant

$$\therefore \frac{U_1}{U_2} = \frac{C_1 V^2}{C_2 V^2} = \frac{e_3 A \times V^2}{e_1 A \times V^2}$$

$$\Rightarrow \frac{U_1}{U_2} = \frac{e_3}{e_1} \Rightarrow \frac{U_1}{U_2} = \frac{15}{10}$$

$$\Rightarrow \left(\frac{U_1}{U_2} - 1\right) \times 100$$

$$= \left(\frac{15}{10} - 1\right) \times 100 = 50\%$$

144. [240] Equivalent capacitance, C_{eq}

$$= \left[\frac{e_0 (7 \times 4)}{4/10} + \frac{5e_0 (1 \times 4)}{4/10} \right] \times 10^{-12}$$

$$C_{eq} = 1.2e_0$$

$$\text{Electrostatic energy} = \frac{1}{2} C_{eq} V^2$$

$$= \frac{1}{2} (1.2e_0) (20) (20) = 240e_0$$

145. [6] Let the final p.d across the combination of capacitor be V . From the conservation of charge

$$C \times 18 + 3C \times 18 = 9C \times V + 3C \times V$$

$$(\therefore K=9)$$

$$72 CV = 12 CV \Rightarrow V = 6 V$$

146. (a) We know that, $Q_0 = CV_0$ [C is capacitance, V_0 is voltage]

Energy stored in the capacitor initially

$$U_i = \frac{1}{2} CV_0^2$$

$$q_1 = \frac{Q_0 C}{C + C} = \frac{2Q_0}{3}, q_2 = \frac{Q_0 C}{C + C} = \frac{Q_0}{3}$$

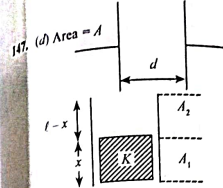
Energy stored in the capacitor finally

$$U_f = \frac{q_1^2}{2C} + \frac{q_2^2}{2C} = \frac{CV_0^2}{3}$$

[By putting value of q_1 and q_2]

Energy loss in the process = $U_i - U_f$

$$= \frac{1}{2} CV_0^2 - \frac{CV_0^2}{3} = \frac{CV_0^2}{6}$$



Before inserting slab, $C_i = \frac{e_0 \epsilon_0 W}{d}$

After inserting dielectric slab $C_f = C_i + C_2$

$$C_f = \frac{K e_0 A \times W}{d} + \frac{e_0 A_2}{d}$$

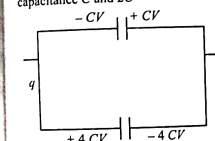
$$C_f = \frac{K e_0 W x}{d} + \frac{e_0 W (t - x)}{d}$$

$$C_f = \frac{2e_0 \epsilon_0 W}{d} = 2C_i$$

$$\therefore \frac{U_i}{U_2} = 2$$

$$\therefore 4x + t - x = 2t \Rightarrow x = \frac{t}{3}$$

148. (a) Here, we have given two capacitor of capacitance C and 2C



$$V_{\text{common}} = \frac{4CV - CV}{3C} = V$$

$$C_{eq} = C + 2C = 3C$$

$$\therefore V_f = \frac{1}{2} \times (3C) V^2 = \frac{3}{2} CV^2$$

$$149. (b) U = \frac{1}{2} CV^2 = \frac{1}{2} \times (12 \times 10^{-12}) \times (10^3)^2$$

$$U = 6 \times 10^{-10} \text{ J}$$

$$U_f = U = 6 \times 10^{-10} \text{ J}$$

$$U_f = \frac{U}{k} = \frac{6 \times 10^{-10}}{6.5}$$

$$U_f = 0.0923 \times 10^{-10}$$

$$\text{Work done } W = U_f - U_i$$

$$W = 6 \times 10^{-10} - 0.0923 \times 10^{-10}$$

150. (a) Work done = change in potential energy

$$\Delta W = U_f - U_i$$

$$= \frac{q^2}{2} \left(\frac{1}{C_1} - \frac{1}{C_2} \right)$$

$$= \frac{(5 \times 10^{-4})^2}{2} \left(\frac{1}{2 \times 10^{-4}} - \frac{1}{5 \times 10^{-4}} \right)$$

$$= 3.75 \times 10^{-4} \text{ J}$$

151. [100] Initially, charge on C_1 , $q_1 = C_1 V$

$$= 5 \times 10^{-6} \times 30 = 150 \mu \text{C}$$

After connecting with C_2 , according to the conservation of charge,

$$\text{Total charge} = 150 \mu \text{C}$$

$$\Rightarrow q_1 + q_2 = 150 \mu \text{C} \quad \dots(i)$$

At equilibrium,

Voltage across C_1 = Voltage across C_2

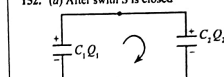
$$\frac{q_1}{C_1} = \frac{q_2}{C_2} \Rightarrow \frac{q_1}{5} = \frac{q_2}{10}$$

$$\Rightarrow q_1 = 2q_2 \quad \dots(ii)$$

$$q_1 = 100 \mu \text{C}$$

On solving (i) and (ii), we get

152. (a) After switch S is closed



$$Q_1 + Q_2 = C_1 V$$

using kirchhoff's voltage law

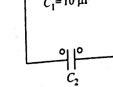
$$\frac{Q_1}{C_1} - \frac{Q_2}{C_2} = 0$$

$$\frac{Q_1}{C_1} = \frac{Q_2}{C_2} \quad \dots(ii)$$

from (i) and (ii)

$$\frac{Q_1}{C_1} + \frac{Q_1}{C_1} = C_1 V \Rightarrow Q_1 = \left(\frac{C_1}{C_1 + C_2} \right) V$$

$$153. (a) \frac{500}{C_1 + 10 \mu F} = \frac{500}{C_2}$$



$$\text{Here } C_1 = 10 \mu F$$

By charge conservation,

$$(50) (10) + 0 = (20) (10) + (20) C_2$$

$$\Rightarrow 500 = 20(10 + C_2)$$

$$\Rightarrow 25 = 10 + C_2 \Rightarrow C_2 = 15 \mu F$$

154. [16] Let charge on C_2 be $q \mu \text{C}$.

$$\text{Therefore, } \frac{q}{2} = \frac{2 \times 10 - q}{2} \Rightarrow q = 16$$

155. (c) According to conservation of charge

$$Q_1 + Q_2 = 12 \mu \text{C} - 3 \mu \text{C} = 9 \mu \text{C}$$

$$\text{As } V_1 = V_2$$

$$\Rightarrow \frac{Q_1}{2R} = \frac{Q_2}{R} \quad [Q \rightarrow Q]$$

$$\Rightarrow Q_1 = 2Q_2$$

$$\Rightarrow Q_1 = \frac{Q_2}{2}$$

[Q_1 and Q_2 are the charges on spheres after connected by a conducting wire]

$$\Rightarrow 2Q_2 - Q_2 = 9 \mu \text{C} \Rightarrow Q_2 = 3 \mu \text{C}$$

$$\text{Hence } Q_1 = 6 \mu \text{C}$$

156. [4] $C_1 = 5 \mu F$, $V_1 = 220 \text{ Volt}$

$$C_2 = 2.5 \mu F$$

$$\text{Heat lost, } \Delta H = U_i - U_f$$

$$= \frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} (V_1 - V_2)^2$$

$$= \frac{1}{2} \times \frac{5 \times 2.5}{5 + 2.5} (220 - 0)^2 = 4 \times 10^{-2} \text{ J}$$

$$= \frac{4}{100} \text{ J}$$

Hence, value of X is 4.

Note: Given answer by JEE Main is x = 36

157. (a) $\Delta H = \Delta U$

$$= \frac{1}{2} \frac{C \times 3C}{C + 3C} (V^2)$$

$$= \frac{1}{2} \times \frac{3}{4} C V^2 = \frac{3}{8} C V^2$$

$$\Delta H = \frac{3}{8} C V^2$$